

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

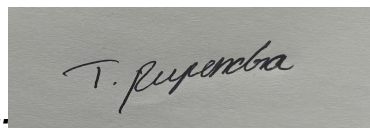
Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

*I **T. Rupendra** / (192311325) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer

Application	20	Correctly	Applies	Limited or	No or
of Knowledge		applies concepts to solve the question/problem	concepts with minor mistakes	partially correct application	incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Assessment Tool 3 - Short Answer Test

① Responsive web Design

Ans:- Responsive web design is an approach to designing websites that automatically adjust their layout, size and content according to the screen size of the device.

Importance:-

- i) Works on mobile, tablet & desktop
- ii) Improves user experience
- iii) Avoids horizontal scrolling
- iv) Provides better accessibility
- v) Reduces need for separate websites

CSS techniques

- i) Media Queries - Apply different styles for different screen sizes.
- ii) Flexible layouts:- uses %, em, rem, vm, and instead of fixed sizes.
- iii) Flexbox:- Arranges elements in rows or columns.
- iv) CSS Grid:- Creates responsive rows & columns
- v) Responsive images - use max-width: 100%.

Example:-

```
<!DOCTYPE html>
<html>
<head>
<style>
  container {
    display: flex;
    gap: 20px;
  }
  bar {
    flex: 1;
    background: light blue;
  }
```

```
@media (max-width: 600px) {
  container {
    flex-direction: column;
  }
}
</style>
</head>
```

Best practices:- use readable fonts, touch-friendly buttons, responsive images, proper spacing & test the website on different devices

2) Form Validation & JavaScript Events.

client-side validation checks user input in the browser the form is submitted to the server.

Purpose:-

- i) prevent invalid data.
- ii) Give immediate feedback.
- iii) Reduce unnecessary server requests.
- iv) improve user experience

JavaScript event handling

JavaScript can use events such as:

- Submit
- Click
- Input
- Change
- Blur

The Submit event can be validate the form before sending data to the server.

Example:-

```
<form id = "my form">
```

email :

```
<input type = "text" id = "email">
```

phone:

```
<input type = "text" id = "phone">
```

```
<button type = "submit"> submit </button>
```

```
</form>  
</script>
```



```
document.getElementById("myform").addEventListener("submit", function(e) {
    let email = document.getElementById("email").value;
    let phone = document.getElementById("phone").value;
    let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    let phonePattern = /^[0-9]{10}$/;
```

Role of Regular Expressions

Regular expressions (Regex) define patterns for validating input.

Examples:-

- email → checks name@domain.com format.
- phone → checks whether the input contains exactly

Improvements:-

- show error messages near the input field
- use HTML5 attributes such as required, type="email" & pattern.
- Do server-side validation also, because client-side validation alone is not secure.
- Sanitize user input to prevent attacks such as XSS.

③ CSS Box model, positioning and media queries

Ans: The CSS box model describes how every HTML element is represented as a rectangular box

Components

- i) content
- ii) padding
- iii) border
- iv) margin

CSS Positioning

Static: Normal document flow

Relative: Moves relative to its normal position.

Absolute: Positioned relative to a positioned

Fixed: Remains fixed relative to the browser

Sticky: Behaves relatively until a specified scroll.

Causes of layout breaking

- fixed widths and height.
- Excessive margins / padding.
- overflowing images.
- Incorrect positioning
- poor use of float
- Lack of responsive CSS.

Media queries:- Media queries apply CSS based on screen size

```
container {
```

```
width: 80%;
```

```
}
```

```
@media (max-width: 600px) {
```

```
  container {
```

```
    width: 100%;
```

```
  }
```

```
}
```

Best practices

- i) use responsive units such as % rem, vw
- ii) use flex box / grid
- iii) use box-sizing: border-box
- iv) use media queries
- v) make images responsive

④ Dom and Dynamic web content.

Ans:- Dom (Document object model) is a programming representation of an HTML document as a tree of objects / elements

Example:-

HTML

+ --- Head

+ --- Body

+ --- H1
 + --- P
 + --- Button

Role of DOM

The DOM allows JavaScript to:

- i) Access HTML elements
- ii) change text and style
- iii) Add or remove elements
- iv) Change attributes
- v) Handle user events

JavaScript DOM manipulation example

```

<h2 id = "title">welcome</h2>
< button onclick = "changeText()" > click me/button

```

```

<script>

```

```

function changeText () {

```

```

document.getElementById ("title").textContent

```

```

= "welcome to IP!";

```

```

}

```

```

</script>

```

When the button is clicked, JavaScript changes the heading without reloading the complete web.

Common DOM method:-

```

document.getElementById ("id");
document.getElementById ("class");
element.textContent = "newText";

```



```
element.style.color = "red";  
element.style.color = ["Hello", "Hello"];
```

Performance & UX improvements

- update only the required DOM elements.
- Avoid unnecessary DOM manipulation.
- use event delegation for many elements.
- use efficient selectors
- minimize repeated layout / reflow operation
- Give immediate visual feedback to user.

Conclusion: DOM manipulation with JavaScript enables dynamic, interactive webpage without required a full page reload.